

Eco-Risk Alpha: Measuring the Green Discount in Counterparty CVA

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1 Introduction

In recent years, the growing emphasis on Environmental, Social, and Governance (ESG) criteria has raised a critical question in financial markets: do firms with stronger ESG performance genuinely benefit from lower credit risk? Furthermore, if such a relationship exists, does it ultimately translate into reduced Credit Valuation Adjustment (CVA) charges applied by banks in over-the-counter (OTC) derivatives trading?

Within the framework of the Summer School “Data Science for Sustainable Finance and Economics” organized by HTW University of Berlin, the project investigates these questions by analyzing a benchmark portfolio consisting of a 5-year payer Interest Rate Swap (IRS). A dataset of 60 corporate entities is employed to construct both statistical and machine learning models explaining 5-year CDS spreads as a function of companies’ sector and regional classifications, fundamental financial indicators, and ESG scores. Observed and model-implied CDS spreads are subsequently transformed into hazard rates, survival probabilities, and, ultimately, into CVA figures. The comparison between market-based and model-based CVA values provides a measure of the potential “green premium.”

The synthetic yet realistic dataset includes company identifiers, sector and regional tags, ESG aggregate and pillar scores (E, S, G), carbon intensity, as well as financial fundamentals such as market capitalization, total assets, debt-to-equity ratio, EBITDA margin, and leverage ratio, alongside observed 5-year CDS spreads. For the CVA computation, the dataset further provides an Expected Exposure (EE) profile of the IRS and a term structure of EUR zero rates (used to compute discount factors), under the assumption of a Loss-Given-Default (LGD) of 60%.

This study is guided by three central research questions:

1. What is the marginal contribution of ESG variables to CDS spreads, and is it statistically and economically significant?
2. To what extent can this marginal contribution influence the computation of CVA?
3. Is the green premium, if present, a systematic effect across firms, or rather a heterogeneous phenomenon?

The methodological framework proceeds in four stages: (i) data cleaning and variable harmonization; (ii) a baseline OLS regression with heteroskedasticity-robust standard errors to isolate the ESG effect; (iii) interpretable machine learning models (Random Forest and Gradient Boosting) to capture non-linearities and interactions; and (iv) the translation of CDS spreads into CVA, with a comparison between observed and predicted values.

2 Theory and Design

The theoretical foundation of this study lies at the intersection of credit risk modeling and sustainable finance. The central objective is to evaluate whether sustainability indicators, specifically ESG scores, provide incremental information about firms' creditworthiness beyond traditional financial fundamentals.

2.1 Credit Default Swaps and CDS Spreads

A Credit Default Swap (CDS) is a financial derivative that allows an investor to swap or offset their credit risk with that of another investor. A protection buyer buys a CDS from a protection seller, who agrees to reimburse them if the borrower defaults [1]. The CDS spread, typically quoted in basis points, reflects the annualized cost of insuring against default risk and serves as a market-based measure of creditworthiness.

2.2 Credit Valuation Adjustment (CVA)

CVA measures the market value of counterparty credit risk embedded in a derivative contract. Although the concept has long been recognized, banking regulations following the 2008 financial crisis made CVA a mandatory element in pricing and risk management. CVA represents the discounted expected loss due to potential counterparty default over the life of a derivative transaction.

Formally, CVA is defined as follows [2]:

$$CVA = (1 - R) \int_0^T EE(t) DF(t) \lambda(t) S(t) dt,$$

where:

- R is the recovery rate, with $R = 1 - LGD$ and $LGD = 0.6$ by assumption,
- $EE(t)$ is the expected exposure at time t ,
- $DF(t)$ is the discount factor at time t ,
- $\lambda(t)$ is the default intensity (hazard rate),
- $S(t)$ is the survival probability at time t ,
- T is the final maturity.

In practice, this integral is approximated by:

$$CVA \approx LGD \cdot \sum_{i=1}^N EE_i \cdot DF_i \cdot (S_{i-1} - S_i),$$

with

$$\lambda = \frac{\text{CDS Spread}}{1 - R}, \quad S_i = \exp(-\lambda \cdot t_i).$$

This formulation establishes an indirect but fundamental link between CDS spreads and CVA.

2.3 Modeling Framework

Two complementary modeling strategies are employed:

(1) Baseline econometric model. An Ordinary Least Squares (OLS) regression with White’s heteroskedasticity-robust standard errors is used to identify the marginal contribution of ESG factors relative to firm size, leverage, and profitability measures. OLS provides interpretability and serves as a benchmark specification.

(2) Machine learning models. Random Forest and Gradient Boosting are implemented to capture non-linear relationships and interactions among predictors. Although less transparent than linear models, these approaches can reveal complex patterns in the data that traditional econometric models may fail to detect.

3 Empirical Study and Testing

3.1 Data Preparation and Cleaning

The raw dataset comprises 60 anonymized firms with sector and region classifications, ESG metrics (aggregate and pillars), carbon intensity, financial fundamentals, and 5-year CDS spreads. The preprocessing steps were as follows:

- **Standardization of categorical labels.** Aliases and capitalization were harmonized (e.g., “Tech” → “Technology”, “north america” → “North America”).
- **Treatment of missing values.** Six missing values were identified in the ESG score. Whenever the aggregate ESG score was unavailable, it was reconstructed as:

$$ESG = 0.5E + 0.3S + 0.2G.$$

Additional missing values were detected in `Carbon_Intensity` and `Market_Cap` (five cases). These were imputed using the sector-wise median. Firms with excessive missing fields were removed (only one observation was discarded).

- **Duplicates.** Duplicate records were identified and removed, retaining a single representative entry.
- **Outliers and skewness.** Outliers were observed in `Market_Cap` and `Carbon_Intensity`. These were deemed *transform* $\log(1 + x)$ was applied.

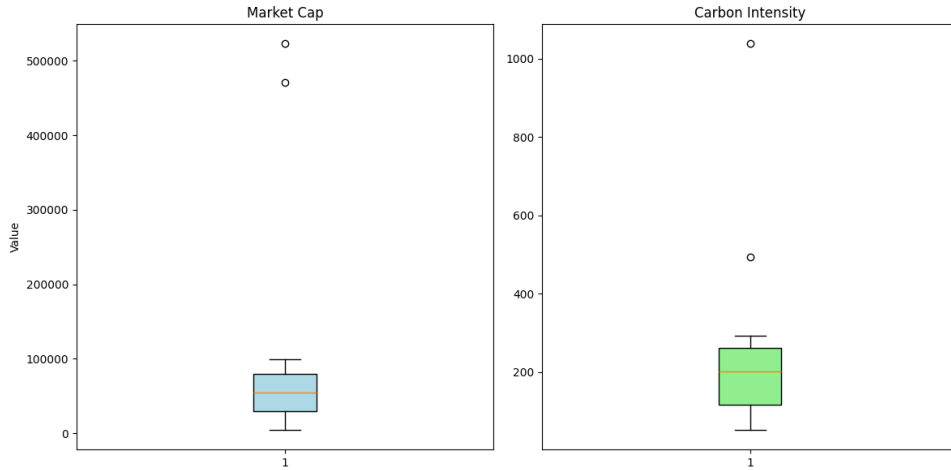


Figure 1: Outlier detection via boxplots for Market Capitalization and Carbon Intensity.

3.2 Exploratory Data Analysis

Following data cleaning, the exploratory analysis investigated linear and non-linear relationships among the variables.

The first step was to visualize a scatter plot between ESG scores and CDS spreads. As shown in Figure 2, the scatter cloud does not suggest the presence of either linear or non-linear dependence between the two variables.

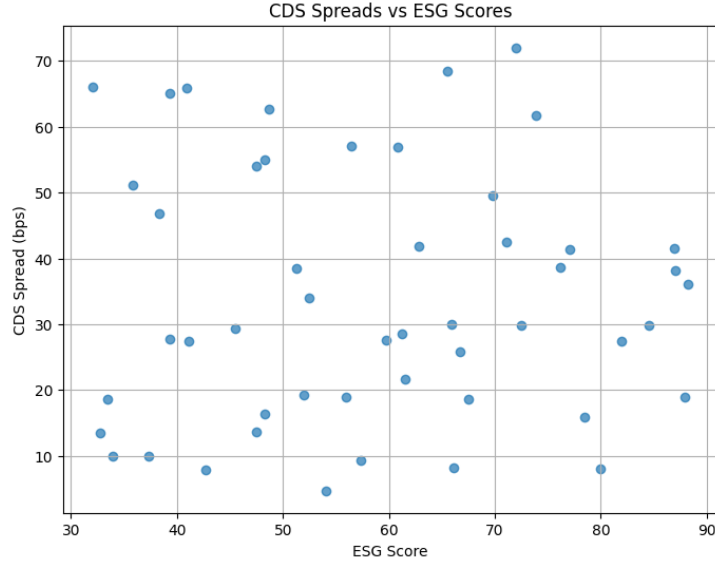


Figure 2: Scatter plot of ESG scores versus CDS spreads.

A pairplot matrix was subsequently generated to analyze all variable interactions. The scatter matrix confirmed the absence of evident linear relationships between CDS spreads and not only with ESG score but with all the explanatory variables in general. This outcome can be attributed both to the large variation in scale and to the inherently non-linear nature of the relationships.

These preliminary findings suggest that linear models such as OLS are unlikely to capture significant patterns, whereas non-linear approaches such as Random Forest and Gradient Boosting may be better suited to uncover relevant signals.

Finally, correlation matrices were computed using both Pearson and Spearman coefficients. Pearson correlation captures linear dependencies, while Spearman correlation is more robust for non-linear relationships and outliers. As illustrated in Figures 3 and 4, both correlation matrices display similar patterns, with only weak associations across features.

The target variable (5-year CDS spread) shows stronger correlations with EBITDA_Margin, Leverage_Ratio, and log_Market_Cap, whereas the ESG_Score exhibits virtually no correlation (Pearson coefficient ≈ 0 , Spearman coefficient ≈ 0.04).

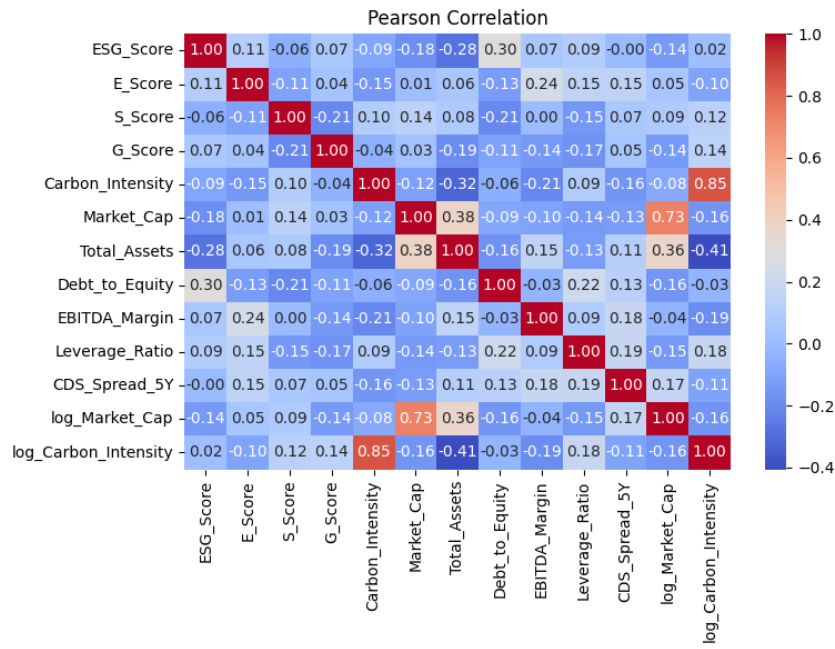


Figure 3: Pearson correlation matrix.

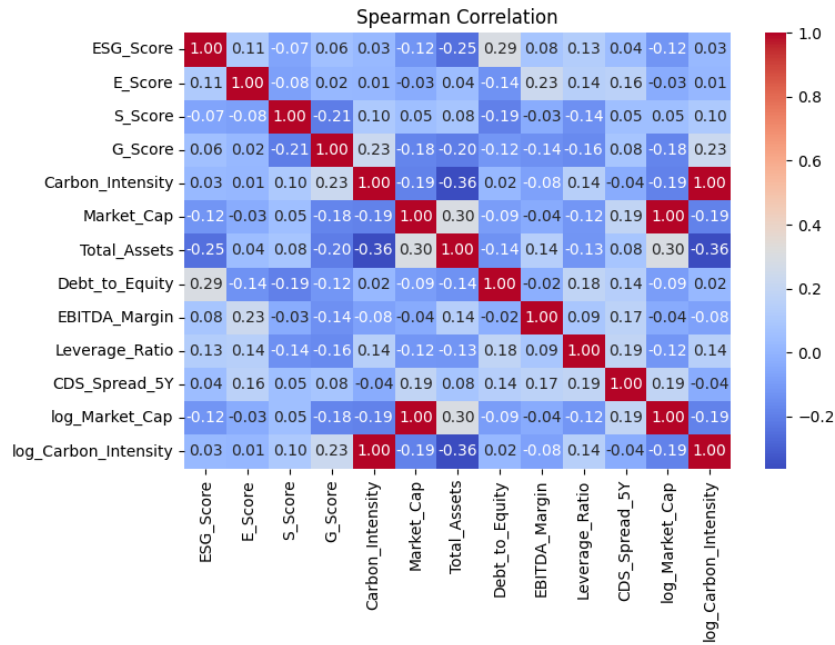


Figure 4: Spearman correlation matrix.

3.3 Econometric Baseline: OLS Regression

To establish a benchmark, a parsimonious linear specification was estimated using a subset of explanatory variables selected to mitigate collinearity concerns.

The chosen regressors were *ESG_Score*, *Total_Assets*, *Leverage_Ratio*, and *EBITDA_Margin*. The estimated model is:

$$\text{CDS}_i = \alpha + \beta_1 \text{ESG_Score}_i + \beta_2 \text{Total_Assets}_i + \beta_3 \text{Leverage_Ratio}_i + \beta_4 \text{EBITDA_Margin}_i + \varepsilon_i,$$

with heteroskedasticity-robust (White’s HC0) standard errors.

As anticipated from the exploratory scatter plots, the regression results confirm the absence of a meaningful linear relationship between the covariates and CDS spreads. The model exhibits very weak explanatory power and none of the regressors are statistically significant at the 5% level.

Key regression statistics are reported below:

Variable	Coefficient	p-value
ESG_Score	0.0034	0.983
Total_Assets	4.1e-05	0.496
Leverage_Ratio	5.8857	0.112
EBITDA_Margin	0.2503	0.249
Model Fit		
R^2	0.075 (Adj. R^2 = -0.007)	
RMSE	18.20	
MAE	14.79	
Cross-validated R^2	-0.123 ± 0.072	
Cross-validated RMSE	19.61 ± 1.64	
Cross-validated MAE	15.95 ± 1.36	

Table 1: OLS Regression: Key Statistics

Interpretation. The linear model offers interpretability but limited explanatory and predictive power; fundamentals dominate, while the estimated marginal ESG effect is weak and statistically fragile.

In conclusion, the OLS regression fails to capture meaningful variation in CDS spreads. The poor explanatory power and lack of statistical significance suggest that alternative specifications, possibly involving different covariates or non-linear methods, are required to have a better model.

3.4 Machine Learning Models

To capture non-linearities and interactions, two ensemble algorithms were considered: Random Forest (RF) and Gradient Boosting (GB). For both models, the same set of explanatory variables was employed: *ESG_Score*, *log_Carbon_Intensity*, *Leverage_Ratio*, *log_Market_Cap*, and *EBITDA_Margin*, in addition to sector and region dummies introduced via one-hot encoding.

Random Forest (RF). The Random Forest model was trained with the following hyperparameters: `n_estimators = 400`, `max_depth = 3`, `random_state = 42`, `n_jobs = -1`, `max_features = 0.6`, `min_samples_leaf = 6`.

The cross-validated (5-fold) performance metrics were:

$$R^2 = -0.073 \pm 0.161, \quad \text{RMSE} = 19.04 \pm 1.14, \quad \text{MAE} = 15.63 \pm 0.56,$$

while the out-of-sample predictions (via `cross_val_predict`) yielded:

$$R^2 = -0.017, \quad \text{RMSE} = 19.08, \quad \text{MAE} = 15.63.$$

The feature importance values (Gini-based) are reported in Table 3.4. Results indicate that firm fundamentals dominate the model, with *log_Market_Cap*, *Leverage_Ratio*, and *EBITDA_Margin* ranking highest, whereas *ESG_Score* exhibits only marginal explanatory power.

Feature	Importance
log_Market_Cap	0.3816
Leverage_Ratio	0.2045
EBITDA_Margin	0.1765
log_Carbon_Intensity	0.1029
ESG_Score	0.0943
Sector_Finance	0.0114
Sector_Technology	0.0096
Region_Europe	0.0079
Region_South America	0.0067
Sector_Healthcare	0.0028
Region_North America	0.0019
Sector_Energy	0.0000
Sector_Industrials	0.0000
Sector_Utillities	0.0000

Table 2: Random Forest Feature Importance

Individual conditional expectation and Partial dependence plot further confirm that ESG_Score related effects on CDS spread are essentially flat and negligible.

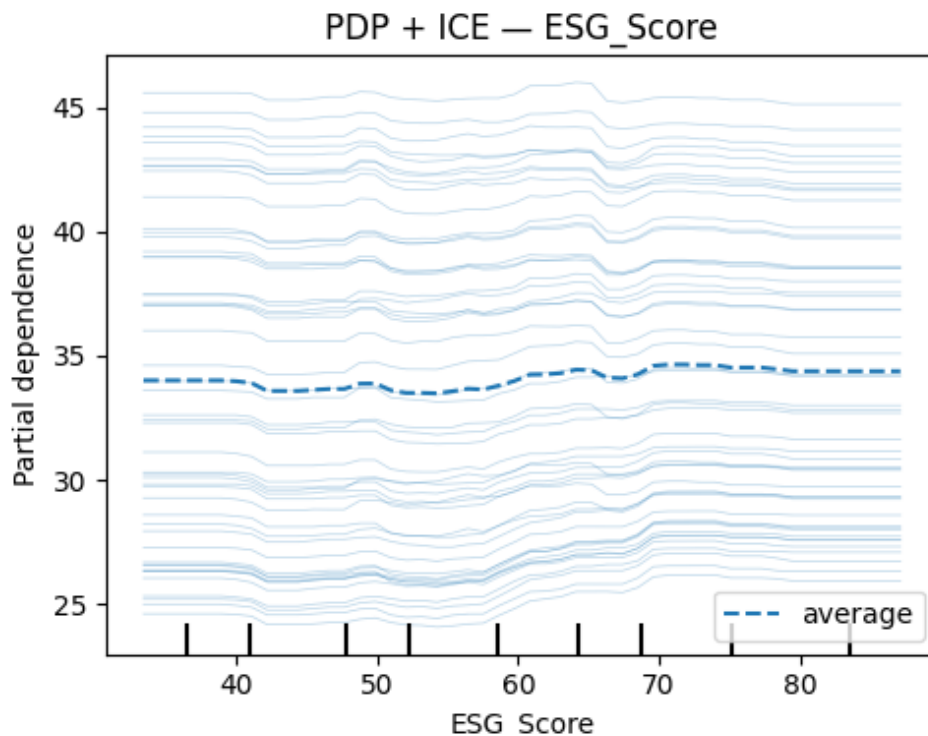


Figure 5: PDP and ICE plot for RF.

Gradient Boosting (GB). The Gradient Boosting model was trained with the following hyperparameters: `n_estimators = 600`, `learning_rate = 0.01`, `max_depth = 3`, `min_samples_leaf = 5`, `random_state = 42`.

The cross-validated (5-fold) performance metrics were:

$$R^2 = -0.325 \pm 0.420, \quad \text{RMSE} = 20.77 \pm 2.18, \quad \text{MAE} = 16.49 \pm 2.05,$$

while the out-of-sample predictions (via `cross_val_predict`) yielded:

$$R^2 = -0.219, \quad \text{RMSE} = 20.89, \quad \text{MAE} = 16.49.$$

The feature importance values (Gini-based) are reported in Table 3.4. Results are broadly consistent with the Random Forest model: firm fundamentals (in particular *log_Market_Cap* and *Leverage_Ratio*) dominate, whereas the ESG_score ranks much lower in explanatory relevance.

Feature	Importance
log_Market_Cap	0.3722
Leverage_Ratio	0.2910
log_Carbon_Intensity	0.1320
EBITDA_Margin	0.1216
ESG_Score	0.0320
Region_North America	0.0300
Region_South America	0.0170
Sector_Finance	0.0038
Sector_Technology	0.0004
Sector_Energy	0.0000
Sector_Industrials	0.0000
Sector_Healthcare	0.0000
Region_Europe	0.0000
Sector_Utillities	0.0000

Table 3: Gradient Boosting Feature Importance

As with the Random Forest, Individual conditional expectation and Partial dependence plot further confirm that ESG-related effects on CDS spreads are essentially flat and negligible.

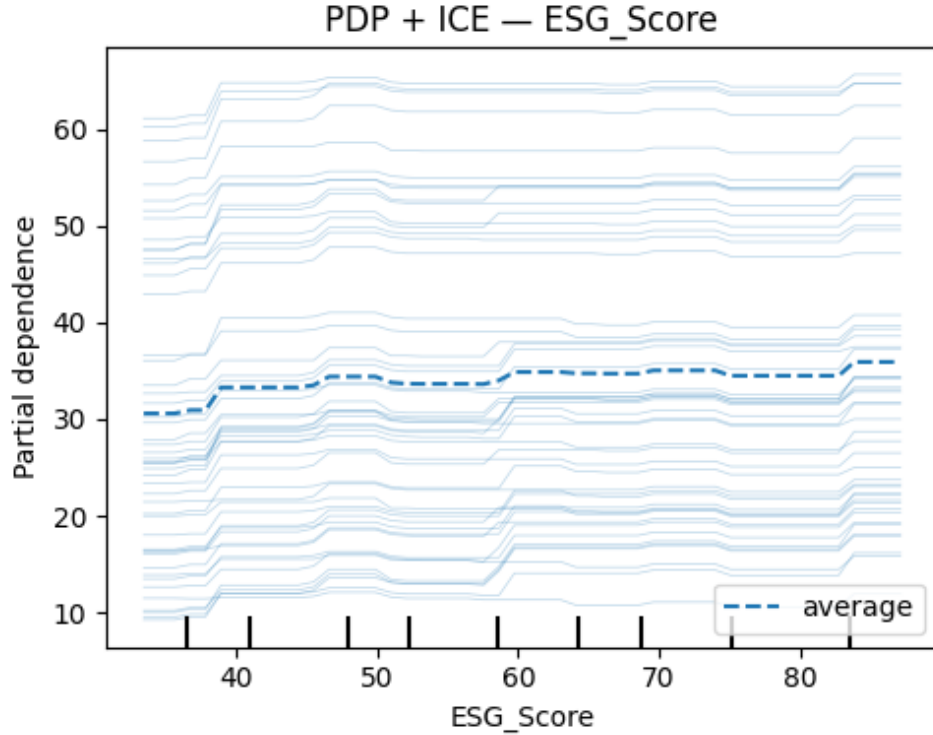


Figure 6: PDP and ICE plot for GB.

3.5 CVA Computation

After performing a piecewise-linear interpolation of the zero rates (so as to obtain a discount curve with values every 0.5 years), the Unilateral CVA for the 5-year payer IRS was computed in three main steps:

1. Mapping CDS spreads to hazard rates:

$$\lambda = \frac{\text{CDS spread}}{1 - R}.$$

2. Computing survival probabilities: On the time grid $\{t_i\}$, survival probabilities are given by:

$$S_i = \exp(-\lambda \cdot t_i).$$

3. Discounting and summation: Discount factors were obtained via continuous compounding, $DF(t) = e^{-rt}$, and the CVA was approximated through the discrete formula:

$$\text{CVA} \approx LGD \cdot \sum_{i=1}^N EE_i \cdot DF_i \cdot (S_{i-1} - S_i).$$

This procedure yields three series of values:

- $\text{CVA}_{\text{actual}}$: computed from observed market CDS spreads,
- CVA_{RF} : computed using Random Forest model-implied CDS spreads,
- CVA_{GB} : computed using Gradient Boosting model-implied CDS spreads.

3.6 Green Premium: Findings and Discussion

The green premium is defined as the difference between market-based and model-implied CVA:

$$\text{Green Premium} = \text{CVA}_{\text{actual}} - \text{CVA}_{\text{predicted}}$$

Starting from this definition, the CVA values implied by the Random Forest (RF) and Gradient Boosting (GB) models ($\text{CVA}_{\text{predicted}}$) were compared with the actual CVA derived from observed CDS spreads ($\text{CVA}_{\text{actual}}$).

Figure 7, presents the distributions of the Green Premium in absolute terms. In both RF and GB cases, the histograms are approximately centered on zero, with mean and median values fluctuating close to zero. This indicates that there is no systematic positive or negative shift in the CVA attributable to ESG-related effects.

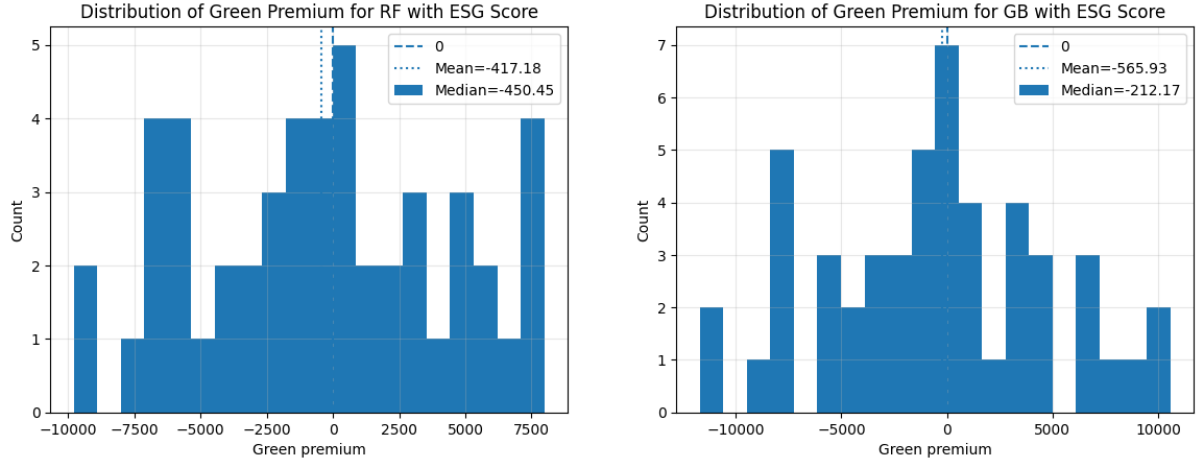


Figure 7: Distribution of the Green Premium in absolute terms.

To account for differences in scale across firms, the distributions were also normalized by dividing ΔCVA by the average of all the actual CVAs. The results, shown in Figure 8, again display values tightly centered around zero. The bounds of the histograms are close to ± 1 , implying that any potential premium is economically negligible.

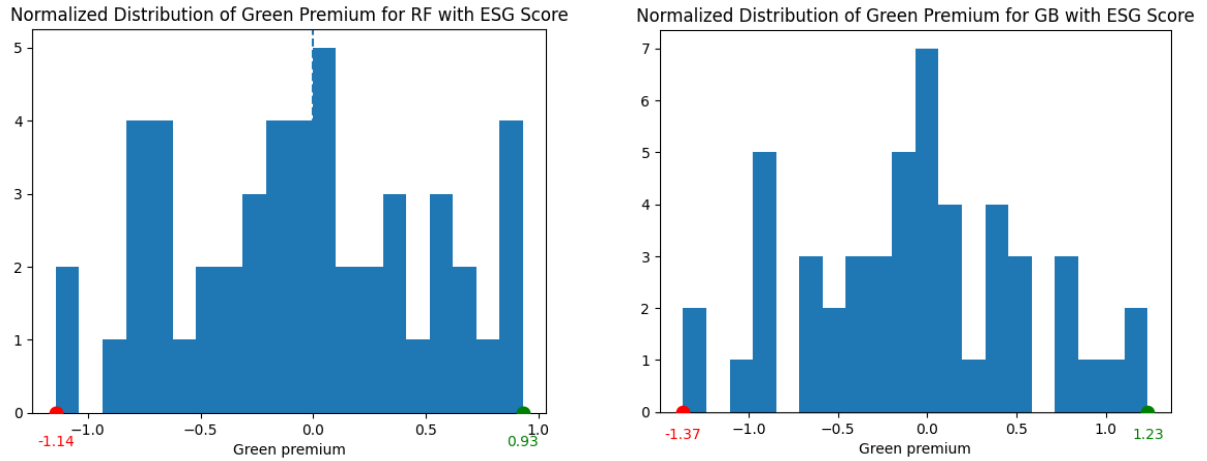


Figure 8: Distribution of the Green Premium in relative terms.

Overall, the analysis consistently indicates the absence of a systematic green premium. Both in absolute and relative terms, the distributions are heterogeneous across firms but remain centered on zero. This outcome is fully consistent with the diagnostics of the predictive models: ESG-related variables had weak marginal importance, and consequently their incremental effect on CVA is negligible.

4 Conclusion

This study set out to investigate whether firms with stronger ESG performance benefit from lower credit risk and whether this translates into a measurable reduction in Credit Valuation Adjustment (CVA).

Using a synthetic dataset of approximately 60 firms, econometric and machine learning approaches were applied to model CDS spreads and map them into CVA values. The empirical evidence does not support the existence of a uniform or systematic “green premium.” Across both linear and non-linear models, balance sheet fundamentals such as leverage and firm size consistently emerged as the dominant drivers of CDS spreads. By contrast, ESG scores displayed weak and non-monotonic effects. When translating spreads into CVA, differences between actual and model-implied values were small and heterogeneous across firms. From a practical perspective, these findings suggest that, in the setting considered, the pricing of counterparty credit risk does not systematically reward ESG performance. CVA adjustments remain primarily determined by traditional financial fundamentals rather than sustainability scores.

Several limitations must be acknowledged. The dataset was synthetic and cross-sectional, with a relatively small sample size. This inevitably constrained statistical power and may have hindered the detection of subtle ESG effects.

In conclusion, the analysis provides a structured framework for linking ESG metrics to CDS spreads and to CVA, but it does not yield conclusive evidence of a systematic green premium. Instead, it highlights the predominance of financial fundamentals in credit risk and underscores the need for richer, higher-quality data to assess whether sustainability is truly priced into counterparty risk adjustments or not.

References

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